SPRINT 1 · DATA ANALYSIS PROJECT

# Airbnb Market Insights

End-to-end data pipeline and dashboard for Airbnb prices in 10 European cities

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## 1. Executive summary

This project builds a complete data workflow on **51,707 Airbnb listings** from 10 European cities: merging 20 source files, enriching them with licensed web data, cleaning them, loading them into a **SQL Server Bronze / Silver / Gold pipeline**, normalizing them into a relational model, and presenting the results in an interactive **Power BI dashboard**.

| 51,689 | 10 | 309,154 | 516 |
|---|---|---|---|
| clean listings | cities | scraped listings for enrichment | neighbourhoods |

**Main findings:** city is by far the biggest price driver (Amsterdam costs about **5x** more per guest than Athens); within a city, **size and location** set the price while guest ratings barely do; professional hosts charge **11%** more with lower guest satisfaction; and groups get much better value (per-guest price falls **44%** from 2 to 5 guests).

## 2. Data sources

| Source | Content | Use |
|---|---|---|
| **Kaggle:** Airbnb Prices in European Cities | 20 CSV files (10 cities x weekdays/weekends), 51,707 listings, 20 columns: price, room type, capacity, ratings, distances, attraction/restaurant indexes, coordinates | Core dataset |
| **Inside Airbnb** (CC BY 4.0) | Detailed listings (309,154) and neighbourhood boundaries for the same 10 cities | Extra features: reviews, ratings, superhosts, amenities, availability |

**Why not scrape Airbnb directly?** Airbnb's Terms of Service prohibit bots and scrapers, and the Kaggle rows have no listing IDs to match against Airbnb pages. Inside Airbnb publishes the same kind of data under a Creative Commons licence, so it was used instead. Selenium was used to collect its download links.

## 3. Data pipeline




| Step | What was done |
|---|---|
| 01 Load & merge | Appended the 20 files; city and day type taken from file names before merging. Row count verified (51,707). |
| 02 Data quality | No missing values, no duplicates. Strong right skew in price (mean EUR 280 vs median EUR 211, max EUR 18,545). Weak correlations with price (only attraction index above 0.3). |
| 03 Web scraping | Selenium opened the Inside Airbnb page and collected the latest files for the 10 cities. |
| 04 Enrichment | Each listing assigned to a neighbourhood by its coordinates (99.9% inside a boundary, rest matched to the nearest, max 809 m). Eight neighbourhood features added. |
| 05 Bronze | Raw data loaded into SQL Server with pandas + SQLAlchemy, with load-tracking columns. |
| 06 Silver | Cleaning (below), Excel export, Silver tables. |
| 07 EDA | Eight business questions answered with charts. |
| Gold (SQL) | Normalized relational schema with primary, foreign and CHECK constraints. |

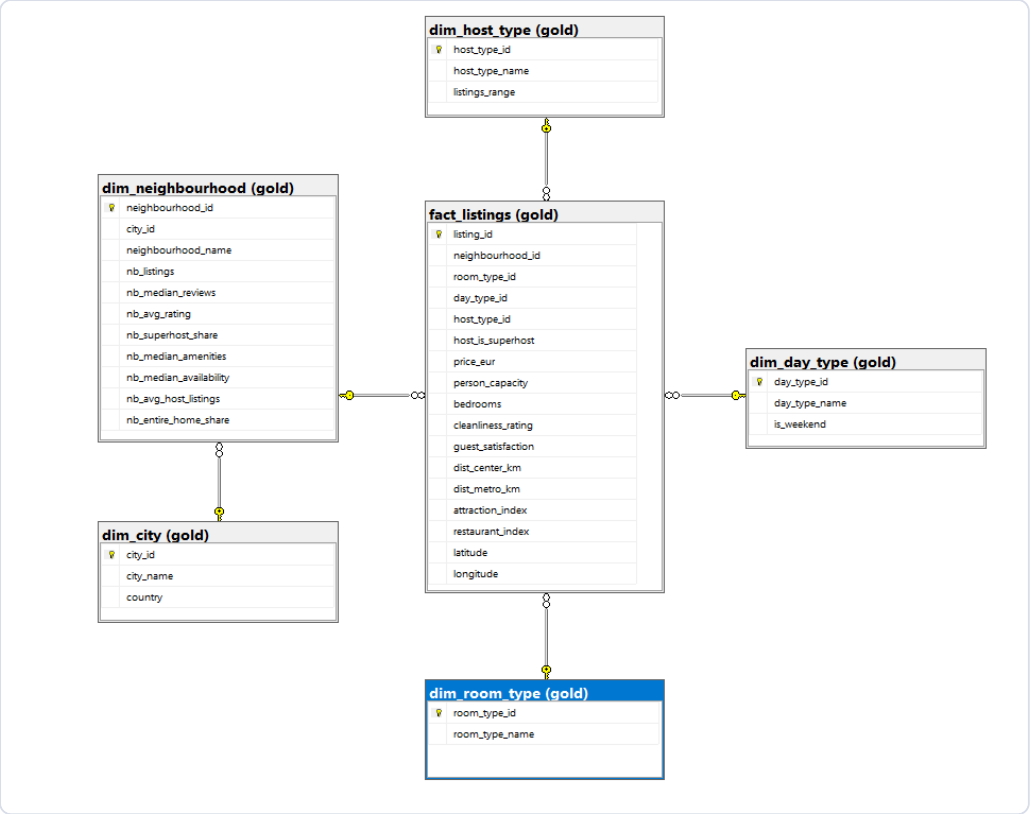
## 4. Data cleaning decisions

| Issue | Decision | Reason |
|---|---|---|
| Extreme prices | Removed 18 listings above 30x their city median | Implausible values (e.g. private rooms for 2 above EUR 12,000). Listings between 20x and 30x kept because they include real luxury homes. |
| Redundant columns | Dropped `room_shared`, `room_private`, raw indexes | Proven to repeat `room_type` exactly; avoids redundancy and multicollinearity. |
| Data types | Capacity as integer, host flags as boolean | Consistent, correct types. |
| Unclear names | e.g. `realSum` → `price_eur`, `biz` → `host_has_over_4_listings` | Readable in SQL and Power BI. |
| Missing neighbourhood features | Left empty (9 rows) | 0.02% of rows; filling would invent data. |
| Greek/Hungarian names | Text stored as `NVARCHAR` | `VARCHAR` turned Greek names into "???", which would merge different neighbourhoods. |

The median is used instead of the mean throughout, because prices are strongly skewed.

## 5. Data model (Gold layer)

The Silver table was normalized into one fact table and five dimensions. Text that repeated thousands of times (city, room type, host type) is stored once and referenced by ID. The city is reached through the neighbourhood rather than stored on the fact table (third normal form).




| Table | Rows | Key |
|---|---|---|
| fact_listings | 51,689 | PK listing_id; FK to neighbourhood, room type, day type, host type |
| dim_neighbourhood | 516 | PK neighbourhood_id; FK city_id |
| dim_city / dim_room_type / dim_day_type / dim_host_type | 10 / 3 / 2 / 3 | Primary keys |

## 6. Dashboard

The Power BI report connects directly to the Gold layer (Import mode). Relationships were detected from the foreign keys. Measures are written in DAX (median price, median price per person, superhost share, weekend premium, price index), and city, room type and day type filters apply to the whole page.




## 7. Key insights

1. **City drives price most.** Median price per guest ranges from EUR 37 (Athens) to EUR 184 (Amsterdam), about 5x. Comparisons must be made within each city.
2. **Location matters inside a city.** Listings within 1 km of the centre cost 1.15x the city median; beyond 8 km they cost 0.70x. In London the centre costs 2.4x the median.
3. **Size and location set the price, not ratings.** Within cities, capacity and closeness to attractions correlate with price (0.4 to 0.6); guest satisfaction and cleanliness barely do.
4. **Mixing cities hides patterns (Simpson's paradox).** Superhosts look cheaper overall (EUR 197 vs 217) but price the same as their city (index 1.02 vs 0.99), because they are more common in cheaper cities.
5. **Professional hosts charge more for less.** Hosts with 5+ listings price 11% above the city median, with lower guest satisfaction (90.1 vs 94.5) and fewer superhosts.
6. **Groups get better value.** Per-guest price drops from EUR 92 (2 guests) to EUR 52 (5 guests), 44% less.
7. **Weekend premiums are small** except in Amsterdam (+14.3%) and Budapest (+8.5%).

## 8. Limitations and next steps

- The Inside Airbnb snapshot is more recent than the Kaggle data, so its features describe neighbourhoods, not individual listings.

- No dates or booking data: occupancy, revenue and seasonality cannot be measured.
- Correlation does not prove causation; explanations above are hypotheses.
- **Next:** a price prediction model (e.g. gradient boosting per city), and more dashboard pages (map, neighbourhood ranking).

**Tools:** Python (pandas, NumPy, Matplotlib, GeoPandas, Selenium, SQLAlchemy) · SQL Server · Power BI (DAX) · Excel

Repository: github.com/KarimElroby7/Airbnb-Europe-Analytics · Data: Kaggle; Inside Airbnb (CC BY 4.0)